Decision-Making Under Uncertainty: Exam Solutions

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Learning objective

Apply minimax regret and Expected Monetary Value (EMV) to select an alternative under uncertainty, and use sensitivity analysis to determine how a subjective probability changes the preferred alternative.

Criteria assessment

Each problem is worth 2 points. Show all calculations, including the complete regret table for C5, every EMV calculation for C6, and the threshold equation and decision ranges for C7. Partial credit may be awarded when the correct procedure is clearly shown but arithmetic errors are made. No points will be awarded if no procedure is shown. A **minimum of 4 points** is required to pass each criterion. A 1-point penalty will be applied to each criterion if the student's name is not written on the assessment or if the assessment is submitted late.

Question	C5	C6	C7
1	2	–	–
2	2	–	–
3	2	–	–
4	–	2	–
5	–	2	–
6	–	2	–
7	–	–	2
8	–	–	2
9	–	–	2
Total	**6**	**6**	**6**

Evaluated criteria

C5 Applies the Minimax Regret criterion by identifying the best payoff in each state, constructing the complete regret table, and selecting the alternative with the smallest maximum regret.

C6 Applies the Expected Monetary Value (EMV) or Bayes criterion by checking the state probabilities, calculating every alternative's expected payoff, and justifying the alternative with the highest EMV.

C7 Uses probability sensitivity analysis to determine remaining probabilities and feasible domains, derive EMV and threshold equations, and identify complete decision intervals including ties.

1 Artisan Apiary Expansion [C5-2]

An artisan apiary must choose one of three hive-expansion plans. Annual net benefits are in thousands of dollars.

	Alternatives		
	Large	Medium	Small
Excellent	76	58	38
Average	34	49	43
Poor	-14	23	35

Questions/Tasks

C5 What decision would be selected using the Minimax Regret criterion? Justify your answer. **[2]**

C5 - Solution [2]

Here A_i follows the order of alternatives in the exam: Large, Medium, Small. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The best payoff in each state is

$$b(S_1) = \max\{76, 58, 38\} = 76, \quad b(S_2) = \max\{34, 49, 43\} = 49, \quad b(S_3) = \max\{-14, 23, 35\} = 35.$$

Using $r(A_i, S_j) = b(S_j) - P_{ay}(A_i, S_j)$ gives

	S_1	S_2	S_3	Maximum regret
A_1	$76 - 76 = 0$	$49 - 34 = 15$	$35 - (-14) = 49$	49
A_2	$76 - 58 = 18$	$49 - 49 = 0$	$35 - 23 = 12$	18
A_3	$76 - 38 = 38$	$49 - 43 = 6$	$35 - 35 = 0$	38

Thus $\min\{49, 18, 38\} = 18$, so the Minimax Regret criterion selects A_2, the medium expansion.

2 Regional Theater Capacity [C5-2]

A theater operator is choosing a seating-capacity plan. Seasonal profit is in thousands of dollars; no reliable probabilities are available.

Alternatives	High	Normal	Low
Add two seating sections	108	40	-28
Add one seating section	82	61	14
Lease a nearby hall	59	52	34
Current seating	34	34	34

Questions/Tasks

C5 What decision would be selected using the Minimax Regret criterion? Justify your answer. **[2]**

C5 - Solution [2]

Here A_i follows the order of alternatives in the exam: add two seating sections, add one seating section, lease a nearby hall, current seating. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The statewise best payoffs are $b(S_1) = 108$, $b(S_2) = 61$, and $b(S_3) = 34$.

	S_1	S_2	S_3	Maximum regret
A_1	$108-108$ $=0$	$61-40$ $=21$	$34-(-28)$ $=62$	62
A_2	$108-82$ $=26$	$61-61$ $=0$	$34-14$ $=20$	26
A_3	$108-59$ $=49$	$61-52$ $=9$	$34-34$ $=0$	49
A_4	$108-34$ $=74$	$61-34$ $=27$	$34-34$ $=0$	74

Since $\min\{62, 26, 49, 74\} = 26$, the recommended decision is A_2 : add one seating section.

3 Emergency Shelter Staffing [C5-2]

A relief organization must select an emergency-shelter staffing plan. Net community value is in thousands of dollars.

	Alternatives		
	Full	Flexible	Limited
Very high	96	74	48
High	54	64	51
Moderate	24	49	46
Low	-24	12	38

Questions/Tasks

C5 What decision would be selected using the Minimax Regret criterion? Justify your answer. **[2]**

C5 - Solution **[2]**

Here A_i follows the order of alternatives in the exam: Full, Flexible, Limited. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The best-payoff vector is

$$(b(S_1), b(S_2), b(S_3), b(S_4)) = (96, 64, 49, 38).$$

	S_1	S_2	S_3	S_4	Maximum regret
A_1	$96-96$ $=0$	$64-54$ $=10$	$49-24$ $=25$	$38-(-24)$ $=62$	62
A_2	$96-74$ $=22$	$64-64$ $=0$	$49-49$ $=0$	$38-12$ $=26$	26
A_3	$96-48$ $=48$	$64-51$ $=13$	$49-46$ $=3$	$38-38$ $=0$	48

For example, $r(A_1, S_4) = 38-(-24) = 62$ and $r(A_3, S_2) = 64-51 = 13$. The smallest maximum regret is $\min\{62, 26, 48\} = 26$, so select A_2, the flexible staffing plan.

4 Coffee Roastery Product Mix [C6-2]

A coffee roastery is selecting a product mix. Profits are in thousands of dollars.

		Strong	Stable	Weak
A_i	$P(S_j)$	0.45	0.35	0.20
Single-origin		86	44	-12
Balanced		64	58	22
Classic		42	47	37

Questions/Tasks

C6 What decision would be selected using the EMV or Bayes criterion? Justify your answer. **[2]**

C6 - Solution **[2]**

Here A_i follows the order of alternatives in the exam: Single-origin, Balanced, Classic. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

First, $0.45 + 0.35 + 0.20 = 1$. Then

$$EMV(A_1) = (0.45)(86) + (0.35)(44) + (0.20)(-12) = 38.7 + 15.4 - 2.4 = 51.7,$$
$$EMV(A_2) = (0.45)(64) + (0.35)(58) + (0.20)(22) = 28.8 + 20.3 + 4.4 = 53.5,$$
$$EMV(A_3) = (0.45)(42) + (0.35)(47) + (0.20)(37) = 18.9 + 16.45 + 7.4 = 42.75.$$

The highest EMV is $\max\{51.7, 53.5, 42.75\} = 53.5$. Therefore, choose A_2, the balanced product mix, with expected profit \$53,500.

5 Aquarium Program Scale [C6-2]

An aquarium is deciding the scale of a temporary education program. Payoffs are in thousands of dollars.

		High Expected		Low
A_i	$P(S_j)$	0.30	0.40	0.30
Major		98	38	-24
Standard		73	53	18
Compact		52	47	34
Regular tours only		27	27	27

Questions/Tasks

C6 What decision would be selected using the EMV or Bayes criterion? Justify your answer. **[2]**

C6 - Solution **[2]**

Here A_i follows the order of alternatives in the exam: Major, Standard, Compact, Regular tours only. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The probabilities are valid because $0.30 + 0.40 + 0.30 = 1$.

$$EMV(A_1) = (0.30)(98) + (0.40)(38) + (0.30)(-24) = 29.4 + 15.2 - 7.2 = 37.40,$$
$$EMV(A_2) = (0.30)(73) + (0.40)(53) + (0.30)(18) = 21.9 + 21.2 + 5.4 = 48.50,$$
$$EMV(A_3) = (0.30)(52) + (0.40)(47) + (0.30)(34) = 15.6 + 18.8 + 10.2 = 44.60,$$
$$EMV(A_4) = (0.30)(27) + (0.40)(27) + (0.30)(27) = 27.00.$$

Because 48.50 is the highest EMV, the Bayes decision is A_2, the standard program, with expected payoff \$48,500.

6 Wind-Farm Maintenance Staffing [C6-2]

A wind-farm operator is choosing a maintenance staffing plan. Payoffs are in thousands of dollars.

		Surge	Strong	Steady	Slow
A_i	$P(S_j)$	0.20	0.30	0.35	0.15
Rapid hiring		105	64	24	-32
Moderate hiring		79	68	48	12
Current staff		53	55	51	42

Questions/Tasks

C6 What decision would be selected using the EMV or Bayes criterion? Justify your answer. **[2]**

C6 - Solution [2]

Here A_i follows the order of alternatives in the exam: Rapid hiring, Moderate hiring, Current staff. States S_j follow the exam order. All payoffs, regrets, and EMVs below are in thousands of dollars unless converted explicitly.

The probability check is $0.20 + 0.30 + 0.35 + 0.15 = 1$.

$$EMV(A_1) = (0.20)(105) + (0.30)(64) + (0.35)(24) + (0.15)(-32) = 21 + 19.2 + 8.4 - 4.8 = 43.8,$$
$$EMV(A_2) = (0.20)(79) + (0.30)(68) + (0.35)(48) + (0.15)(12) = 15.8 + 20.4 + 16.8 + 1.8 = 54.8,$$
$$EMV(A_3) = (0.20)(53) + (0.30)(55) + (0.35)(51) + (0.15)(42) = 10.6 + 16.5 + 17.85 + 6.3 = 51.25.$$

Thus $\max\{43.8, 54.8, 51.25\} = 54.8$, so choose A_2, moderate hiring, with expected payoff \$54,800.

7 Concert Merchandise Order [C7-2]

A concert vendor must choose between a large and a small merchandise order. Payoffs are in thousands of dollars.

A_i	High (S_1)	Low (S_2)
Large	80	20
Small	50	40

Questions/Tasks

C7 Let $P(S_1) = p$. State the preferred alternative for every probability value ($0 \le p \le 1$). **[2]**

C7 - Solution [2]

The listed states are assumed mutually exclusive and exhaustive, so their probabilities sum to one. Thus $P(S_2) = 1 - p$. Nonnegative probabilities require $p \ge 0$ and $1 - p \ge 0$, giving the feasible domain $0 \le p \le 1$.

$$EMV(A_1) = 80p + 20(1 - p) = 20 + 60p,$$
$$EMV(A_2) = 50p + 40(1 - p) = 40 + 10p.$$

At the threshold, the EMVs are equal:

$$EMV(A_2) = EMV(A_2) \quad \longrightarrow \quad 20 + 60p = 40 + 10p \quad \longrightarrow \quad 50p = 20 \quad \longrightarrow \quad p = \frac{2}{5} = 0.4$$

Evaluating both EMVs at $p = 0$ and $p = 1$:

$$p = 0: \quad EMV(A_1) = 20 < 40 = EMV(A_2) \longrightarrow A_2$$
$$p = 1: \quad EMV(A_1) = 80 > 50 = EMV(A_2) \longrightarrow A_1$$

$$p < 0.4 \longrightarrow A_2, \qquad p = 0.4 \longrightarrow \text{Tie}, \qquad p > 0.4 \longrightarrow A_1.$$

$$\begin{cases} A_2, & 0 \le p < \frac{2}{5}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{2}{5}, \\ A_1, & \frac{2}{5} < p \le 1. \end{cases}$$

8 Meal-Kit Launch [C7-2]

A meal-kit company is comparing a bold launch with a measured launch. Payoffs are in thousands of dollars.

	High	Moderate	Low
A_i	S_1	S_2	S_3
Bold	60	30	0
Measured	35	45	30

Questions/Tasks

C7 Suppose $P(S_1) = p$, $P(S_2) = \frac{p}{3}$. State the preferred alternative for every feasible probability value of p. **[2]**

C7 - Solution [2]

The listed states are assumed mutually exclusive and exhaustive, so their probabilities sum to one. Thus

$$P(S_3) = 1 - p - \frac{p}{3} = 1 - \frac{4p}{3}.$$

Nonnegative probabilities require $p \ge 0$, $\frac{p}{3} \ge 0$, and $1 - \frac{4p}{3} \ge 0$, giving the feasible domain $0 \le p \le \frac{3}{4}$.

$$EMV(A_1) = 60p + 30\left(\frac{p}{3}\right) + 0\left(1 - \frac{4p}{3}\right) = 70p,$$

$$EMV(A_2) = 35p + 45\left(\frac{p}{3}\right) + 30\left(1 - \frac{4p}{3}\right) = 30 + 10p.$$

At the threshold, the EMVs are equal:

$$EMV(A_1) = EMV(A_2) \quad \longrightarrow \quad 70p = 30 + 10p \quad \longrightarrow \quad 60p = 30 \quad \longrightarrow \quad p = \frac{1}{2} = 0.5$$

Evaluating both EMVs at $p = 0$ and $p = \frac{3}{4}$:

$$p = 0: \quad EMV(A_1) = 0 < 30 = EMV(A_2) \longrightarrow A_2$$

$$p = \frac{3}{4}: \quad EMV(A_1) = 52.5 > 37.5 = EMV(A_2) \longrightarrow A_1$$

$$p < 0.5 \longrightarrow A_2, \qquad p = 0.5 \longrightarrow \text{Tie}, \qquad p > 0.5 \longrightarrow A_1.$$

$$\begin{cases} A_2, & 0 \le p < \frac{1}{2}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{1}{2}, \\ A_1, & \frac{1}{2} < p \le \frac{3}{4}. \end{cases}$$

9 Publishing-Platform Bid [C7-2]

A publishing firm must choose between a standard platform bid and an experimental platform bid. Payoffs are in thousands of dollars.

	Immediate	Revision	Delayed
A_i	S_1	S_2	S_3
Standard	30	70	70
Experimental	50	50	30

Questions/Tasks

C7 Suppose $P(S_1) = 3p$, $P(S_2) = p$. State the preferred alternative for every feasible probability value of p. **[2]**

C7 - Solution [2]

The listed states are assumed mutually exclusive and exhaustive, so their probabilities sum to one. Thus

$$P(S_3) = 1 - 3p - p = 1 - 4p.$$

Nonnegative probabilities require $3p \geq 0$, $p \geq 0$, and $1 - 4p \geq 0$, giving the feasible domain $0 \leq p \leq \frac{1}{4}$.

$$EMV(A_1) = 30(3p) + 70p + 70(1 - 4p) = 70 - 120p,$$
$$EMV(A_2) = 50(3p) + 50p + 30(1 - 4p) = 30 + 80p.$$

At the threshold, the EMVs are equal:

$$EMV(A_1) = EMV(A_2) \quad \longrightarrow \quad 70 - 120p = 30 + 80p \quad \longrightarrow \quad 40 = 200p \quad \longrightarrow \quad p = \frac{1}{5} = 0.2$$

Evaluating both EMVs at $p = 0$ and $p = \frac{1}{4}$:

$$p = 0: \quad EMV(A_1) = 70 > 30 = EMV(A_2) \longrightarrow A_1$$
$$p = \frac{1}{4}: \quad EMV(A_1) = 40 < 50 = EMV(A_2) \longrightarrow A_2$$

$$p < 0.2 \longrightarrow A_1, \qquad p = 0.2 \longrightarrow \text{Tie}, \qquad p > 0.2 \longrightarrow A_2.$$

$$\begin{cases} A_1, & 0 \leq p < \frac{1}{5}, \\ A_1 \text{ and } A_2 \text{ are tied}, & p = \frac{1}{5}, \\ A_2, & \frac{1}{5} < p \leq \frac{1}{4}. \end{cases}$$

10 Summarized Solutions [C5-C7]

All monetary values below are in thousands of dollars. In each problem, A_i and S_j follow the original alternative and state order; $E_i = EMV(A_i)$ and $\boldsymbol{q} = (P(S_1), \ldots, P(S_n))$. For C5, $r_{ij} = b_j - P_{ay}(A_i, S_j)$, where b_j is the best payoff in state S_j.

10.1 Artisan Apiary Expansion C5–2

A_1: Large; A_2: Medium; A_3: Small hive expansion.
S_1: Excellent; S_2: Average; S_3: Poor.
Annual net benefits; $\boldsymbol{b} = (76, 49, 35)$.

Regret	S_1	S_2	S_3	Max.
A_1	$76 - 76 = 0$	$49 - 34 = 15$	$35 - (-14) = 49$	49
A_2	$76 - 58 = 18$	$49 - 49 = 0$	$35 - 23 = 12$	18
A_3	$76 - 38 = 38$	$49 - 43 = 6$	$35 - 35 = 0$	38

$$\min\{49, 18, 38\} = 18 \quad \Longrightarrow \quad A_2: \text{ Medium expansion; maximum regret 18.}$$

10.2 Regional Theater Capacity C5–2

A_1: Add two seating sections; A_2: Add one seating section;
A_3: Lease a nearby hall; A_4: Current seating.
S_1: High; S_2: Normal; S_3: Low.
Seasonal profit; $\boldsymbol{b} = (108, 61, 34)$.

Regret	S_1	S_2	S_3	Max.
A_1	$108 - 108 = 0$	$61 - 40 = 21$	$34 - (-28) = 62$	62
A_2	$108 - 82 = 26$	$61 - 61 = 0$	$34 - 14 = 20$	26
A_3	$108 - 59 = 49$	$61 - 52 = 9$	$34 - 34 = 0$	49
A_4	$108 - 34 = 74$	$61 - 34 = 27$	$34 - 34 = 0$	74

$$\min\{62, 26, 49, 74\} = 26 \quad \Longrightarrow \quad A_2: \text{ Add one seating section; maximum regret 26.}$$

10.3 Emergency Shelter Staffing C5–2

A_1: Full; A_2: Flexible; A_3: Limited staffing.
S_1: Very high; S_2: High; S_3: Moderate; S_4: Low.
Net community value; $\boldsymbol{b} = (96, 64, 49, 38)$.

Regret	S_1	S_2	S_3	S_4	Max.
A_1	$96-96=0$	$64-54=10$	$49-24=25$	$38-(-24)=62$	62
A_2	$96-74=22$	$64-64=0$	$49-49=0$	$38-12=26$	26
A_3	$96-48=48$	$64-51=13$	$49-46=3$	$38-38=0$	48

$$\min\{62, 26, 48\} = 26 \quad \Longrightarrow \quad A_2: \text{ Flexible staffing; maximum regret 26.}$$

10.4 Coffee Roastery Product Mix C6–2

A_1: Single-origin; A_2: Balanced; A_3: Classic product mix.
S_1: Strong; S_2: Stable; S_3: Weak. Profits in thousands of dollars.

$$\boldsymbol{q} = (0.45, 0.35, 0.20), \qquad 0.45 + 0.35 + 0.20 = 1.$$

$$\begin{aligned} E_1 &= 0.45(86) + 0.35(44) + 0.20(-12) = 51.70, \\ E_2 &= 0.45(64) + 0.35(58) + 0.20(22) = 53.50, \\ E_3 &= 0.45(42) + 0.35(47) + 0.20(37) = 42.75. \end{aligned}$$

A_2: Balanced product mix; EMV 53.50 = \$53,500.

10.5 Aquarium Program Scale C6–2

A_1: Major; A_2: Standard; A_3: Compact; A_4: Regular tours only.
S_1: High; S_2: Expected; S_3: Low. Payoffs in thousands of dollars.

$$\boldsymbol{q} = (0.30, 0.40, 0.30), \qquad 0.30 + 0.40 + 0.30 = 1.$$

$$\begin{aligned} E_1 &= 0.30(98) + 0.40(38) + 0.30(-24) = 37.40, \\ E_2 &= 0.30(73) + 0.40(53) + 0.30(18) = 48.50, \\ E_3 &= 0.30(52) + 0.40(47) + 0.30(34) = 44.60, \\ E_4 &= 0.30(27) + 0.40(27) + 0.30(27) = 27.00. \end{aligned}$$

A_2: Standard program; EMV 48.50 = \$48,500.

10.6 Wind-Farm Maintenance Staffing C6–2

A_1: Rapid hiring; A_2: Moderate hiring; A_3: Current staff.
S_1: Surge; S_2: Strong; S_3: Steady; S_4: Slow. Payoffs in thousands of dollars.

$$\boldsymbol{q} = (0.20, 0.30, 0.35, 0.15), \qquad 0.20 + 0.30 + 0.35 + 0.15 = 1.$$

$$\begin{aligned} E_1 &= 0.20(105) + 0.30(64) + 0.35(24) + 0.15(-32) = 43.80, \\ E_2 &= 0.20(79) + 0.30(68) + 0.35(48) + 0.15(12) = 54.80, \\ E_3 &= 0.20(53) + 0.30(55) + 0.35(51) + 0.15(42) = 51.25. \end{aligned}$$

A_2 : Moderate hiring; EMV 54.80 = \$54,800.

10.7 Concert Merchandise Order C7–2

A_1: Large; A_2: Small merchandise order. S_1: High; S_2: Low.
Payoffs in thousands of dollars; probabilities sum to one and are nonnegative:

$$\boldsymbol{q} = (p, 1-p), \qquad p \geq 0,\ 1-p \geq 0 \quad \Longrightarrow \quad 0 \leq p \leq 1.$$

$$\begin{aligned} E_1 &= 80p + 20(1-p) = 20 + 60p, \\ E_2 &= 50p + 40(1-p) = 40 + 10p. \end{aligned}$$

$$\begin{aligned} E_1 = E_2 &\iff 20 + 60p = 40 + 10p \\ &\iff 50p = 20 \iff p = \tfrac{2}{5} = 0.4. \end{aligned}$$

$$\begin{cases} A_2, & 0 \leq p < \frac{2}{5}, \\ A_1, A_2 \text{ tied}, & p = \frac{2}{5}, \\ A_1, & \frac{2}{5} < p \leq 1. \end{cases}$$

10.8 Meal-Kit Launch C7–2

A_1: Bold; A_2: Measured launch. S_1: High; S_2: Moderate; S_3: Low.
Payoffs in thousands of dollars; probabilities sum to one and are nonnegative:

$$\boldsymbol{q} = \left(p, \tfrac{p}{3}, 1 - \tfrac{4p}{3}\right), \qquad p \geq 0,\ \tfrac{p}{3} \geq 0,\ 1 - \tfrac{4p}{3} \geq 0 \quad \Longrightarrow \quad 0 \leq p \leq \tfrac{3}{4}.$$

$$\begin{aligned} E_1 &= 60p + 30(\tfrac{p}{3}) + 0(1 - \tfrac{4p}{3}) = 70p, \\ E_2 &= 35p + 45(\tfrac{p}{3}) + 30(1 - \tfrac{4p}{3}) \\ &= 30 + 10p. \end{aligned}$$

$$\begin{aligned} E_1 = E_2 &\iff 70p = 30 + 10p \\ &\iff 60p = 30 \iff p = \tfrac{1}{2} = 0.5. \end{aligned}$$

$$\begin{cases} A_2, & 0 \leq p < \frac{1}{2}, \\ A_1, A_2 \text{ tied}, & p = \frac{1}{2}, \\ A_1, & \frac{1}{2} < p \leq \frac{3}{4}. \end{cases}$$

10.9 Publishing-Platform Bid C7–2

A_1: Standard; A_2: Experimental platform bid.
S_1: Immediate; S_2: Revision; S_3: Delayed. Payoffs in thousands of dollars.

$$\boldsymbol{q} = (3p, p, 1-4p), \qquad 3p \geq 0,\ p \geq 0,\ 1-4p \geq 0 \quad \Longrightarrow \quad 0 \leq p \leq \tfrac{1}{4}.$$

$$E_1 = 30(3p) + 70p + 70(1-4p) = 70 - 120p,$$
$$E_2 = 50(3p) + 50p + 30(1-4p) = 30 + 80p.$$

$$E_1 = E_2 \iff 70 - 120p = 30 + 80p$$
$$\iff 40 = 200p \iff p = \tfrac{1}{5} = 0.2.$$

$$\begin{cases} A_1, & 0 \le p < \frac{1}{5}, \\ A_1, A_2 \text{ tied}, & p = \frac{1}{5}, \\ A_2, & \frac{1}{5} < p \le \frac{1}{4}. \end{cases}$$